

Coarsening Confers Robustness: Causal Bayesian Optimization under Graph Misspecification

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Abstract

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1 Introduction

Many scientific problems require choosing interventions in complex systems: which variables to manipulate, and at what values, to optimize a target under a limited experimental budget. Bayesian optimization (BO) fits a probabilistic surrogate to costly evaluations and selects experiments sequentially. Standard BO treats the system as a black box and therefore does not use its causal structure.

Causal Bayesian optimization (CBO) [1] closes this gap by using an assumed causal graph to select intervention variables and observational data to construct prior information about their effects. Existing methods are typically presented with a directed acyclic graph (DAG); in this paper we also allow acyclic directed mixed graphs so that the supplied graph can represent latent confounding. Because causal graphs are elicited from experts or estimated from finite data, the assumed graph can be wrong. Such an error can do more than slow learning: it can exclude the best intervention from the search space.

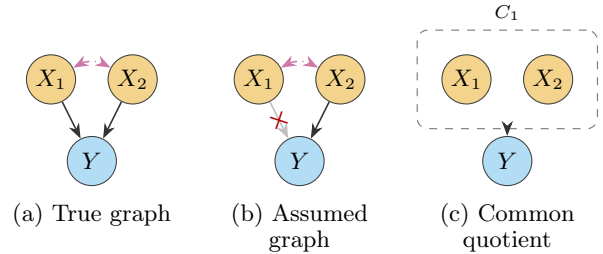


Figure 1: The PARALLELPARENT construction. Deleting $X_1 \rightarrow Y$ changes the fine exploration set but not the quotient because the cluster edge $C_1 \rightarrow Y$ remains through $X_2 \rightarrow Y$. Orange nodes are manipulable, blue is the target, and the dash-dotted edge denotes latent confounding.

Figure 1 illustrates this failure mode. The confounded parents X_1 and X_2 jointly determine Y , and the optimum requires setting both. If the assumed variable-level graph (the *fine graph*) omits $X_1 \rightarrow Y$, CBO discards every intervention containing X_1 .

This observation motivates changing the resolution of the graph supplied to the optimizer. A partition Π groups the variables into disjoint clusters. The resulting cluster-level directed acyclic graph (C-DAG) states relations between clusters while leaving their internal structure unspecified. In Figure 1(c), X_1 and X_2 form the cluster $C_1 = \{X_1, X_2\}$. The edge $C_1 \rightarrow Y$ survives through $X_2 \rightarrow Y$, so the whole-cluster intervention remains available.

We call the resulting method quotient causal Bayesian optimization (QCBO). QCBO receives a partition and its C-DAG and restricts interventions to unions of manipulable clusters. A cluster may encode a known subsystem, policy module, or other domain-defined unit whose external relations are more reliable than its internal structure. We assume that the practitioner provides the partition and C-DAG. Coarsening removes sensitivity to hidden within-cluster distinctions, but it can also remove useful fine-grained actions. It does not protect against quotient-visible errors: deleting both

parent edges in Figure 1 removes $C_1 \rightarrow Y$ and therefore changes the C-DAG consumed by QCBO.

Our contributions are fourfold:

1. We construct quotient versions of arm-based, dynamic, and model-based causal Bayesian optimizers, together with an adaptive refinement variant.
2. We formalize the *quotient contract*: every graph-dependent operation uses the supplied graph only through its quotient. This yields exact trajectory invariance when two fine graphs have the same quotient.
3. We characterize the value lost by restricting interventions to cluster unions, prove monotonicity under refinement, and give structural conditions under which coarsening is lossless.
4. Experiments isolate exploration-set and prior corruption, measure the action-space price, and test portability across optimizer families.

Proofs, complete dataset graphs, and additional results appear in the appendices.

2 Background

2.1 Causal models and interventions

A structural causal model (SCM) $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, F, P(\mathbf{U}) \rangle$ consists of exogenous variables $\mathbf{U}$, endogenous variables $\mathbf{V}$, their structural mechanisms $F = \{f_V : V \in \mathbf{V}\}$, and an exogenous distribution $P(\mathbf{U})$ [10]. Each endogenous variable is assigned by

$$V := f_V(\text{Pa}(V), \mathbf{U}_V),$$

where $\text{Pa}(V)$ are its endogenous parents and $\mathbf{U}_V \subseteq \mathbf{U}$ are its exogenous inputs. We write $\text{An}(V)$ for the ancestors of V .

Let $Y \in \mathbf{V}$ be the target and $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ the manipulable variables. An intervention consists of a nonempty set $A \subseteq \mathbf{X}$ and assigned values $a \in D(A)$, where $D(A)$ is the intervention domain. The operator $\text{do}(A = a)$ replaces the structural mechanisms of the variables in A by the constants a . The resulting interventional response is

$$f_A(a) := \mathbb{E}_{\mathcal{M}}[Y \mid \text{do}(A = a)]. \quad (1)$$

A directed acyclic graph (DAG) represents the directed relations among the endogenous variables when all relevant common causes are observed. When latent common causes are present, the latent projection of the SCM is an acyclic directed mixed graph (ADMG)

$\mathcal{G}$: directed edges represent causal relations and bidirected edges represent latent confounding. Throughout, $\mathcal{G}$ denotes the graph of the true data-generating SCM and $\hat{\mathcal{G}}$ denotes the graph supplied to the optimizer. Changing $\hat{\mathcal{G}}$ can change the optimizer’s decisions, but it does not change the outcomes generated by $\mathcal{M}$.

2.2 Causal Bayesian optimization

Causal Bayesian optimization (CBO) [1] searches for an intervention that minimizes Equation (1). It maintains a Gaussian process (GP) for each retained intervention set, also called an *arm*. Its exploration set consists of the graph’s minimal intervention sets (MIS) [6]. Writing $\hat{\mathcal{G}}_{\bar{A}}$ for the graph obtained by deleting incoming edges to A , this set is

$$\mathbf{ES}(\hat{\mathcal{G}}) = \text{MIS}(\hat{\mathcal{G}}) := \left\{ A \subseteq \mathbf{X} : \begin{array}{l} A \neq \emptyset, \\ A \subseteq \text{An}(Y) \text{ in } \hat{\mathcal{G}}_{\bar{A}} \end{array} \right\}. \quad (2)$$

Thus A is retained only when every intervened variable remains an ancestor of Y after the intervention removes incoming edges to A .

For every retained arm, CBO places a GP prior on its response surface,

$$f_A \sim \mathcal{GP}\left(m_A^{\hat{\mathcal{G}}}, k_A\right), \quad (3)$$

where $m_A^{\hat{\mathcal{G}}}$ and k_A are the mean and covariance functions. When the effect is identified under $\hat{\mathcal{G}}$, the prior mean uses an observational estimate,

$$m_A^{\hat{\mathcal{G}}}(a) := \hat{\mathbb{E}}_{\hat{\mathcal{G}}}[Y \mid \text{do}(A = a)]. \quad (4)$$

Otherwise, our implementation uses the observational mean. Sequential interventions are selected with the base optimizer’s acquisition rule; the cost-normalized expected-improvement rule and the supported causal estimators are recorded in Appendix B.

The supplied graph therefore affects CBO through two objects:

$$\hat{\mathcal{G}} \mapsto \left(\mathbf{ES}(\hat{\mathcal{G}}), \{m_A^{\hat{\mathcal{G}}}\}_{A \in \mathbf{ES}(\hat{\mathcal{G}})} \right).$$

A graph error can change the prior information, the available arms, or both.

2.3 Dynamic and model-based CBO

Dynamic causal Bayesian optimization (DCBO) [2] extends CBO to temporal causal models. It selects interventions within successive time slices while using the slice graphs and temporal transitions to transfer information over time.

Model-based causal Bayesian optimization (MCBO) [11] instead places GP models on the individual structural mechanisms and composes them along the causal graph. It then optimizes interventions using a GP upper confidence bound (GP-UCB) acquisition rule. These families depend on the supplied graph in different ways: DCBO through its temporal graph components and MCBO through its mechanism factorization.

2.4 Cluster causal graphs

Let Π be a partition of $\mathbf{V}$, and let $\chi : \mathbf{V} \rightarrow \Pi$ map each variable to its unique cluster. The quotient graph $\mathcal{G}^\Pi$ has a directed edge $\chi(u) \rightarrow \chi(v)$ whenever $u \rightarrow v$ is present in $\mathcal{G}$ and $\chi(u) \neq \chi(v)$; bidirected edges are mapped analogously. The quotient is a cluster-level causal graph, or C-DAG. A partition is *valid* when the directed part of its quotient is acyclic [8].

A C-DAG preserves which causal and confounding relations exist between clusters while leaving within-cluster structure unspecified. C-DAG identification is sound and complete over the fine graphs represented by the quotient [3]. This established identification layer is the input to the optimizers introduced next.

3 Quotient causal optimizers

Each quotient optimizer replaces only the graph-dependent components of its base method. It receives a partition and a valid supplied C-DAG and otherwise retains the base method’s acquisition and update rules.

3.1 Arm-based quotient CBO

Quotient causal Bayesian optimization (QCBO) receives the partition Π , supplied C-DAG $\hat{\mathcal{G}}^\Pi$, target Y , observational data, and intervention domains. We take every nonmanipulable variable, including Y , to be a singleton cluster. Hence a cluster that contains a manipulable variable contains only manipulable variables. Let

$$\Pi_M := \{C \in \Pi : C \subseteq \mathbf{X}\}$$

be the manipulable clusters, let $C_Y := \chi(Y)$ be the target cluster, and let $U_\Pi(S) := \bigcup_{C \in S} C$ expand a set of clusters to its variables. QCBO applies the MIS rule to the C-DAG:

$$\mathbf{ES}^\Pi(\hat{\mathcal{G}}^\Pi) := \left\{ U_\Pi(S) : \begin{array}{l} \emptyset \neq S \subseteq \Pi_M, \\ S \subseteq \text{An}(C_Y) \text{ in } (\hat{\mathcal{G}}^\Pi)_S \end{array} \right\}. \quad (5)$$

Every retained arm therefore intervenes on complete clusters.

For each arm $A \in \mathbf{ES}^\Pi(\hat{\mathcal{G}}^\Pi)$, a complete causal-effect identification procedure determines whether the observational distribution identifies its prior mean. The

resulting two-tier prior is

$$m_A^\Pi(a) = \begin{cases} \hat{\mathbb{E}}[Y \mid \text{do}(A = a)] & \text{if identified,} \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (6)$$

Non-identifiable effects start from the observational mean and are learned from interventions.

Assumption 1 (Estimator boundary). *The supplied partition is valid. The identification test is complete, while the numerical implementation is sound for the supported backdoor, frontdoor, and g-computation functionals but is not complete. An identified functional outside this implemented class receives the observational-mean prior.*

As a sanity check, the singleton partition recovers the fine graph, the CBO exploration set, and the CBO prior construction arm by arm. A formal statement and proof appear in Appendix A.

3.2 Dynamic and model-based quotient variants

Quotient dynamic CBO (QDCBO) replaces every DCBO time-slice graph and its temporal transitions by their quotients. It selects cluster-union interventions within each slice and otherwise retains DCBO’s temporal acquisition and update logic.

Quotient model-based CBO (QMCBO) replaces MCBO’s node mechanisms by joint conditional cluster mechanisms

$$P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)}),$$

and composes them along the C-DAG. Here $\text{pa}(C)$ denotes the parent clusters of C . Joint rather than marginal mechanisms preserve dependence among variables within a cluster.

An SCM is *causally sufficient* when every common cause of its endogenous variables is represented in the model, so its causal graph has no bidirected edges. Under causal sufficiency and a valid partition, the joint cluster factorization is sound for whole-cluster interventions. It does not determine effects of interventions on strict subsets of a cluster; two fine models can agree on every quotient-level input and disagree on such a sub-cluster effect. Appendix A states and proves both facts.

Family	Graph-dependent quotient object
CBO $\rightarrow$ QCBO	intervention arms and causal priors
DCBO $\rightarrow$ QDCBO	time-slice graphs and transitions
MCBO $\rightarrow$ QMCBO	joint cluster mechanisms

3.3 Adaptive refinement

Hierarchical quotient CBO (HQCBO) begins from a coarse partition and a declared refinement hierarchy. When its plateau rule detects stalled progress, it splits clusters participating in the incumbent. It rejects a split if the resulting directed quotient is cyclic, retains data for arms that remain available, and initializes only newly exposed arms. The plateau rule and hierarchy are heuristic inputs: refinement is not guaranteed to expose a useful fine-grained intervention. Section 4.5 states the conditional recovery and invariance guarantees.

4 Robustness and the price of coarsening

4.1 Quotient invariance

For a graph G and partition Π , define the quotient input

$$Q_\Pi(G) := (\Pi, G^\Pi). \quad (7)$$

For a complete run, let

$$\xi := (\mathcal{M}, \mathcal{D}_0, \omega, \eta)$$

collect the fixed true SCM, initial data, optimizer random stream, and hyperparameters. The true SCM belongs to the environment rather than the optimizer's input. Let $\tau_{\mathcal{A}}(\hat{\mathcal{G}}, \xi)$ denote the complete trajectory of optimizer $\mathcal{A}$: its selected arms, intervention values, observations, and incumbents.

An optimizer satisfies the *quotient contract* when every graph-dependent operation uses $\hat{\mathcal{G}}$ only through $Q_\Pi(\hat{\mathcal{G}})$. For a true graph $\mathcal{G}$, define the protected class

$$E_\Pi(\mathcal{G}) := \{\hat{\mathcal{G}} \text{ ADMG over } \mathbf{V} : Q_\Pi(\hat{\mathcal{G}}) = Q_\Pi(\mathcal{G})\}. \quad (8)$$

Within-cluster edits and quotient-redundant cross-cluster edits remain in this class. An edit that adds or removes a quotient edge does not.

Proposition 1 (Exact quotient invariance). *At fixed ξ , every optimizer satisfying the quotient contract obeys*

$$Q_\Pi(\hat{\mathcal{G}}) = Q_\Pi(\hat{\mathcal{G}}') \implies \tau_{\mathcal{A}}(\hat{\mathcal{G}}, \xi) = \tau_{\mathcal{A}}(\hat{\mathcal{G}}', \xi).$$

Equal quotient inputs give equal first decisions. The fixed true SCM then gives the coupled runs equal observations, so induction yields identical subsequent decisions and observations. In Figure 1, deleting $X_1 \rightarrow Y$ is protected because $X_2 \rightarrow Y$ preserves the quotient edge. Deleting both parent edges is quotient-visible and is not protected.

Corollary 1 (Portability of the quotient contract). *QCBO and QMCBO satisfy Proposition 1; QDCBO satisfies it within every time slice. At the singleton partition, QCBO, QDCBO, and QMCBO reduce to CBO, DCBO, and MCBO, respectively.*

The converse characterization—trajectory invariance over every protected class implies factorization through Q_Π —is stated in Appendix A.

4.2 Price of restricting interventions

The action family allowed by Π consists of nonempty unions of manipulable clusters:

$$\mathcal{U}(\Pi) := \left\{ \bigcup_{C \in S} C : \emptyset \neq S \subseteq \Pi_M \right\}. \quad (9)$$

For an intervention set A , define its best population value by

$$\mu^*(A) := \min_{a \in D(A)} f_A(a).$$

Let $\Pi_{\text{fine}} := \{\{v\} : v \in \mathbf{V}\}$ denote the singleton partition. The best value reachable at resolution Π and its loss relative to the fine action space are

$$V(\Pi) := \min_{A \in \mathcal{U}(\Pi)} \mu^*(A), \quad \Delta(\Pi) := V(\Pi) - V(\Pi_{\text{fine}}). \quad (10)$$

We call $\Delta(\Pi) \geq 0$ the *price of coarsening*.

A partition Π' *refines* Π , written $\Pi' \preceq \Pi$, when every cluster of Π' is contained in a cluster of Π . Refinement exposes additional interventions.

Proposition 2 (Monotonicity and price of coarsening). *For valid $\Pi' \preceq \Pi$,*

$$\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi'), \quad 0 \leq \Delta(\Pi') \leq \Delta(\Pi).$$

Moreover,

$$\Delta(\Pi) = 0 \iff \arg \min_{A \in \mathcal{U}(\Pi_{\text{fine}})} \mu^*(A) \cap \mathcal{U}(\Pi) \neq \emptyset.$$

Thus coarsening is lossless exactly when at least one fine-optimal intervention is a union of clusters. The MIS-reduction result in Appendix A connects this action-family value to the quotient exploration set. Even when $\Delta(\Pi) = 0$, a partition can change arm count, intervention dimension, cost, and finite-budget behavior.

4.3 Protected-class regret separation

Let $\hat{z}_t = (\hat{A}_t, \hat{a}_t)$ be the incumbent intervention after trial t , and define its population value by

$$\mu(\hat{z}_t) := \mathbb{E}_{\mathcal{M}}[Y \mid \text{do}(\hat{A}_t = \hat{a}_t)].$$

With $y^* := V(\Pi_{\text{fine}})$, define instantaneous and cumulative incumbent regret by

$$r_t := \mu(\hat{z}_t) - y^*, \quad R_T := \sum_{t=1}^T r_t. \quad (11)$$

Fix a true SCM $\mathcal{M}$ with graph $\mathcal{G}$, while the optimizer receives a hypothesis graph $\hat{\mathcal{G}}$. Let $R_T(\mathcal{A}, \hat{\mathcal{G}})$ denote regret at fixed ξ , and define worst-case regret over the protected class by

$$R_T^{\text{wc}}(\mathcal{A}, \Pi, \mathcal{G}) := \sup_{\hat{\mathcal{G}} \in E_\Pi(\mathcal{G})} R_T(\mathcal{A}, \hat{\mathcal{G}}).$$

For a cluster-union incumbent,

$$\mu(\hat{z}_t) - y^* = \underbrace{V(\Pi) - y^*}_{\text{coarsening price}} + \underbrace{\mu(\hat{z}_t) - V(\Pi)}_{\text{optimization error}}. \quad (12)$$

A *semi-Markovian SCM* has mutually independent exogenous variables, each of which is a parent of at most two endogenous variables; shared exogenous parents represent the bidirected edges of its latent-projection ADMG.

Theorem 1 (Protected-class regret separation). *The following statements hold.*

- (i) *For every $\beta > 0$, there exist a semi-Markovian SCM, a valid partition with $\Delta(\Pi) = 0$, and a hypothesis $\hat{\mathcal{G}} \in E_\Pi(\mathcal{G})$ obtained by deleting one directed edge, such that every optimizer whose initial and sequential interventions lie in $\text{MIS}(\hat{\mathcal{G}})$ and whose incumbent is selected among executed interventions satisfies*

$$r_t \geq \beta \quad \text{for every } t, \quad R_T^{\text{wc}}(\mathcal{A}, \Pi, \mathcal{G}) \geq \beta T.$$

- (ii) *If an optimizer satisfies Proposition 1, executes only cluster-union interventions, and selects its incumbent among executed actions, then its trajectory is identical for every $\hat{\mathcal{G}} \in E_\Pi(\mathcal{G})$ and*

$$\begin{aligned} R_T^{\text{wc}}(\mathcal{A}, \Pi, \mathcal{G}) &= R_T(\mathcal{A}, \mathcal{G}) \\ &= T\Delta(\Pi) + \sum_{t=1}^T (\mu(\hat{z}_t) - V(\Pi)). \end{aligned}$$

Part (i) is instantiated by the exploration-set corruption experiment in Section 5.2: a quotient-invisible edge deletion can force linear regret even when coarsening is lossless. Part (ii) separates the unavoidable structural price from optimization error within the quotient action space.

4.4 When coarsening is lossless

For a fine intervention $A \subseteq \mathbf{X}$, define its *cluster completion*

$$A^\Pi := \bigcup \{C \in \Pi : C \cap A \neq \emptyset\}.$$

This is the smallest cluster-union intervention containing A .

Definition 1 (Full intervention support). *A partition has full intervention support when, after any fine intervention $\text{do}(A = a)$, every natural value attainable by the additional variables in $A^\Pi \setminus A$ is also permitted when completing the intervention to A^Π .*

Definition 2 (Cluster-contained confounding). *Represent a semi-Markovian SCM by a nonparametric structural equation model with mutually independent exogenous variables. A partition has cluster-contained confounding when, for every $A \subseteq \mathbf{X}$, the exogenous ancestors of the naturally responding variables $A^\Pi \setminus A$ after $\text{do}(A)$ are disjoint from the exogenous ancestors of Y after $\text{do}(A^\Pi)$.*

Corollary 2 (Graphical zero-price condition). *In a semi-Markovian SCM, full intervention support and cluster-contained confounding imply*

$$V(\Pi) = V(\Pi_{\text{fine}}) \quad \text{and} \quad \Delta(\Pi) = 0.$$

These conditions are sufficient but not necessary. The practitioner-facing assumption is the structural placement of confounding. Appendix A introduces the potential-outcome notation needed to derive the corresponding cross-world independence statement and prove the corollary.

4.5 Fixed partitions and refinement

Proposition 3 (Linear regret floor). *For a fixed partition,*

$$r_t \geq \Delta(\Pi) \quad \text{for every } t, \quad R_T \geq T\Delta(\Pi).$$

A positive coarsening price therefore creates a regret floor that no learning within the fixed action space can remove. The adaptive method of Section 3 can remove it only by visiting a sufficiently fine partition.

Definition 3 (Incumbent consistency). *The base optimizer is incumbent-consistent on the action space of a partition Π_J if continued optimization on that space satisfies $\mu(\hat{z}_t) \rightarrow V(\Pi_J)$ almost surely.*

Corollary 3 (Conditional refinement recovery). *If the refinement schedule visits a valid partition Π_J with $\Delta(\Pi_J) = 0$ and the base optimizer is incumbent-consistent on that action space, then*

$$r_t \rightarrow 0, \quad R_T/T \rightarrow 0.$$

Corollary 4 (Adaptive refinement invariance). *Let $\tau_{\mathcal{H}}$ denote the HQCBO trajectory under a declared hierarchy $\mathcal{H}$. If the visited partitions are $\Pi_0, \dots, \Pi_J$, then*

$$[\forall j, \mathcal{G}^{\Pi_j} = (\mathcal{G}')^{\Pi_j}] \implies \tau_{\mathcal{H}}(\mathcal{G}) = \tau_{\mathcal{H}}(\mathcal{G}').$$

Recovery is conditional on the hierarchy exposing a lossless partition, the heuristic trigger reaching it, and the base optimizer being incumbent-consistent. Refinement can also reveal previously hidden graph errors, so validity must be checked after every split.

5 Experiments

5.1 Evaluation design and datasets

The evaluation separates four claims: robustness to exploration-set corruption, robustness to causal-prior corruption, the action-space price of coarsening, and portability of the quotient contract. Table 1 summarizes this design. In every misspecification comparison, the true SCM, initial data, hyperparameters, and random stream are paired. Only the graph supplied to the optimizer changes.

The controlled suite uses three SCMs with closed-form population effects: PARALLEL PARENT, FRONT DOOR, and MEDIATED CHAIN. Their fine graphs, partitions, and quotients appear in Appendix D. Each SCM provides one shared observational dataset; the optimizer receives its first $n_{\text{obs}} = 100$ observations. All displayed disturbance variables in this suite are mutually independent standard Gaussians; shared latent variables induce the stated bidirected edges. Interventional outcomes, including the initial design, are evaluated using the exact population expectation $\mathbb{E}[Y \mid \text{do}(A = a)]$. We use 30 paired seeds, three initial intervention points per arm, unit cost per intervened variable, and $T = 50$ trials, increased to 60 for MEDIATED CHAIN.

The portability suite follows each base method’s released protocol. The CBO family contains ToyGraph, CompleteGraph (“Synthetic”), and SimplifiedCoralGraph, with 10 seeds, $n_{\text{obs}} = 100$, ten initial points per arm, 40 trials, and minimization. The Coral simulator is the linear structural equation model (SEM) benchmark introduced by Aglietti et al. [1]: its graph is based on a Bermuda reef ecosystem model and its mechanisms were fit to 50 field observations. The DCBO family uses stationary, independent, and non-stationary synthetic temporal SEMs, with 20 seeds, three time slices, ten trials per slice, and minimization. The MCBO family uses its ToyGraph and PSA-Graph simulators with 20 seeds, $T = 100$, GP-UCB exploration parameter $\beta = 10$, zero observation noise,

and maximization. PSAGraph is the model-based harness’s adaptation of the prostate-specific-antigen simulator used in the original CBO study.

We report four complementary quantities: the best-so-far population objective, cumulative incumbent regret for the controlled minimization experiments, paired trajectory deviation for testing exact invariance, and the final objective value with its standard error across seeds. The *best-so-far population value* evaluates the incumbent in the true SCM. For the controlled minimization suite, *cumulative incumbent regret* is Equation (11). *Paired trajectory deviation* compares the selected arms, intervention values, and incumbents of coupled correct- and wrong-graph runs. For the portability suite we report the final value as mean $\pm$ standard error.

5.2 Exploration-set corruption

The PARALLEL PARENT SCM is shown in Figure 1 and Appendix D. Its structural equations are

$$\begin{aligned} U &= 0.4\varepsilon_0, \\ X_1 &= U + 0.3\varepsilon_1, \\ X_2 &= U + 0.3\varepsilon_2, \\ Y &= (X_1 - 2)^2 + (X_2 + 2)^2 + 0.1\varepsilon_Y. \end{aligned} \tag{13}$$

Under the correct graph, the fine MIS are $\{X_1\}$, $\{X_2\}$, and $\{X_1, X_2\}$, and the joint arm attains $y^* = 0$ at $(2, -2)$. Deleting $X_1 \rightarrow Y$ from the hypothesis graph removes every arm containing X_1 and leaves only $\{X_2\}$. Its exact population optimum is 4.25, so no amount of optimization under the wrong fine graph can close this gap.

After 50 trials, correct-graph CBO reaches 0.0016 ± 0.0002 , whereas misspecified CBO ends at 4.2500 ± 0.0000 (mean $\pm$ standard error). Their cumulative incumbent regrets are 4.21 ± 0.37 and 216.23 ± 1.12 , respectively. The latter value is close to the structural lower bound $4.25T = 212.5$; the excess is finite-budget optimization error within the misspecified action space.

For $\Pi = \{\{X_1, X_2\}, \{Y\}\}$, the same deletion is invisible because $X_2 \rightarrow Y$ preserves the quotient edge. The paired QCBO runs select the same arms and intervention values at every trial, and their incumbent trajectories have zero deviation. Figure 2(a) therefore separates a permanent fine-graph arm exclusion from exact quotient invariance. A quotient-visible $X_1 \leftrightarrow Y$ perturbation serves as the negative control and changes the QCBO trajectory, with maximum paired incumbent deviation 21.45.

Table 1: Evaluation map. Arrow up for maximization and down for minimization objective give the objective direction.

Study	Data	Perturbation	Comparison	Objective	Primary metric	Purpose
A	PARALLEL PARENT	delete $X_1 \rightarrow Y$	CBO versus QCBO	↓	regret; deviation	exploration-set robustness
B	FRONT DOOR	add $X_1 \leftrightarrow M$	CBO versus QCBO	↓	regret; deviation	prior robustness
C	MEDIATED CHAIN	refine $\{X_1, X_2\}$	CBO, QCBO, HQCBO	↓	value; regret	price and recovery
D	eight family tasks	quotient graph view	base versus quotient	↓ / ↑	final objective value	portability

5.3 Prior corruption without arm loss

The FRONT DOOR graph in Appendix D contains $X_1 \rightarrow M \rightarrow Y$ and latent $X_1 \leftrightarrow Y$. Its SCM is

$$\begin{aligned}
 U &= 0.4\varepsilon_0, \\
 X_1 &= U + 0.3\varepsilon_1, \\
 M &= 2X_1 + 0.2\varepsilon_M, \\
 Y &= 2 \left[1 - \exp \left\{ -\frac{(M-1)^2}{2(0.3)^2} \right\} \right] + U + 0.1\varepsilon_Y.
 \end{aligned} \tag{14}$$

Its population objective is minimized by $\text{do}(M = 1)$. Under the correct graph, $\text{do}(M)$ is identified by adjustment for X_1 and $\text{do}(X_1)$ by the frontdoor criterion. Adding a spurious $X_1 \leftrightarrow M$ leaves the fine MIS unchanged but makes both prior means non-identifiable. This design therefore changes the prior while holding the available arms fixed.

Fine CBO loses its observational head start and accumulates additional regret while searching for the narrow optimum, but it can recover from interventional observations because the optimal arm remains available. The paired increase in cumulative regret is 5.91 ± 1.04 at $T = 50$; the final values are 0.0003 ± 0.0003 under the correct graph and 0.0035 ± 0.0022 under the corrupted graph. This is a transient finite-budget effect rather than an action-space exclusion like in 5.2. The perturbation is internal to $\{X_1, M\}$ and disappears in the quotient, so the paired QCBO decisions and incumbent trajectories are identical. The right panel of Figure 2 contrasts this transient fine-graph effect with exact quotient invariance.

5.4 Action-space price and refinement

The MEDIATED CHAIN, in Appendix D, graph has $X_1 \rightarrow X_2 \rightarrow Y$, with

$$X_2 = 2X_1 + \varepsilon_2, \quad Y = (X_2 - 4)^2 + \varepsilon_Y.$$

Fine CBO can execute $\text{do}(X_1 = 2)$ without intervening on X_2 . The edge $X_1 \rightarrow X_2$ therefore remains active, giving

$$\begin{aligned}
 X_2 &= 4 + \varepsilon_2, \\
 \mathbb{E}[Y \mid \text{do}(X_1 = 2)]_{\text{amp}} &= \text{Var}(\varepsilon_2) = 0.04.
 \end{aligned}$$

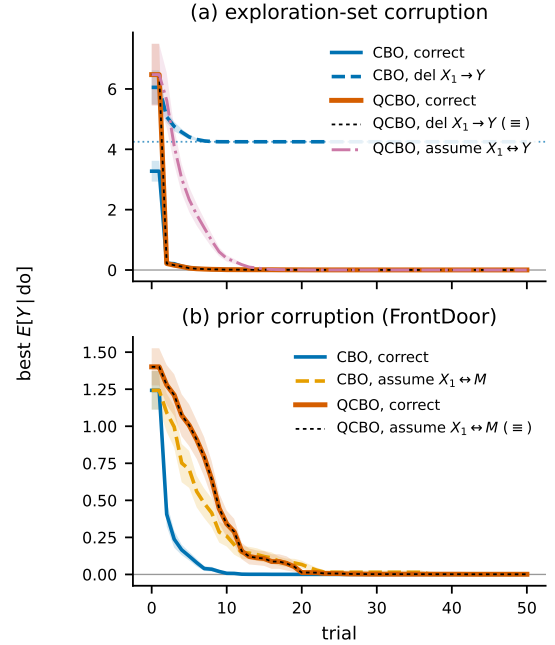


Figure 2: Correct and misspecified trajectories, mean $\pm$ standard error over 30 seeds. Top: deleting $X_1 \rightarrow Y$ removes the optimal fine arm, whereas QCBO is unchanged because the edit is invisible in the quotient. Bottom: corruption of an identified prior slows fine CBO, whereas QCBO remains exactly invariant.

Fixed QCBO cannot execute $\text{do}(X_1)$ alone because X_1 and X_2 belong to the same manipulable cluster. Its only arm intervenes jointly:

$$\text{do}(X_1 = x_1, X_2 = x_2).$$

Intervening on X_2 removes the edge $X_1 \rightarrow X_2$, so x_1 no longer affects the target. Since the intervention domain restricts $x_2 \in [-2, 2]$, the best permitted value is

$$\min_{x_2 \in [-2, 2]} (x_2 - 4)^2 = 4.$$

Hence

$$V(\Pi) = 4, \quad y^* = 0.04, \quad \Delta(\Pi) = 3.96.$$

Consequently, every fixed-QCBO incumbent has instantaneous regret at least 3.96, and Proposition 3

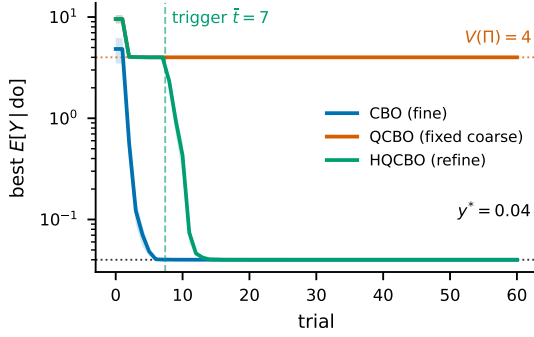


Figure 3: Best-so-far population objective on MEDIATEDCHAIN, mean $\pm$ standard error over 30 seeds. Fixed QCBO cannot execute $\text{do}(X_1)$; refinement exposes this arm.

gives the permanent lower bound $R_T \geq 3.96T$. Fine CBO does not incur this floor because its MIS retains $\{X_1\}$. HQCBO removes it only after the declared hierarchy splits $\{X_1, X_2\}$ and exposes the singleton arm. Figure 3 shows the resulting best-value and cumulative-regret trajectories. CBO, fixed QCBO, and HQCBO finish at 0.0400 ± 0.0000 , 4.0000 ± 0.0000 , and 0.0400 ± 0.0000 , with cumulative regrets 5.46 ± 1.61 , 243.23 ± 0.90 , and 36.97 ± 1.18 , respectively. HQCBO refines at trial 7.4 ± 0.1 ; all 30 declared splits are accepted. The fixed-QCBO result exceeds the theoretical floor $60 \times 3.96 = 237.6$ only by its within-space optimization error. The experiment demonstrates recovery for this declared split; it does not imply that the plateau rule always selects a useful refinement.

5.5 Portability across optimizer families

Figure 4 compares each quotient construction with its base optimizer on the eight tasks described above; all corresponding DAGs and C-DAGs are in Appendix D. The structural prediction is exact: QDCBO is invariant in all 60 paired dynamic runs and QMCBO in all 40 paired model-based runs, whereas their base methods change under the same graph-view perturbations. This confirms that the quotient contract is independent of the family-specific acquisition and surrogate construction.

Table 2 gives the final population value and paired quotient disadvantage for each task. For $\downarrow$ rows the paired difference is quotient minus base; for $\uparrow$ rows it is base minus quotient. Positive values therefore always mean that the quotient ended worse.

Table 2: Final objective value, reported as mean $\pm$ standard error across seeds. The arrows indicate the optimization direction: $\downarrow$ denotes minimization and $\uparrow$ maximization.

Task	Base final	Quotient final
<i>CBO vs. QCBO ($n = 10$ seeds)</i>		
Toy $\downarrow$	-2.16 ± 0.00	-2.13 ± 0.03
Synthetic $\downarrow$	-3.14 ± 0.19	-1.25 ± 0.07
Coral $\downarrow$	36.08 ± 0.02	36.42 ± 0.01
<i>DCBO vs. QDCBO (3 time slices; $n = 20$ seeds)</i>		
Stationary $\downarrow$	-6.14 ± 0.05	-6.43 ± 0.02
Independent $\downarrow$	-3.12 ± 0.06	-5.57 ± 0.15
Nonstationary $\downarrow$	8.03 ± 0.90	6.33 ± 0.29
<i>MCBO vs. QMCBO ($n = 20$ seeds)</i>		
ToyGraph $\uparrow$	1.39 ± 0.13	2.16 ± 0.00
PSAGraph $\uparrow$	-5.15 ± 0.00	-5.15 ± 0.00

6 Related work

Graph-agnostic and causal entropy optimization methods learn over candidate graphs online [5, 9], while causal-bandit guarantees price learning an unknown graph within the horizon [7]. Our setting instead studies commitment to a fixed coarse input. These approaches are complementary: a graph learner could operate over, or trigger refinement of, a quotient.

The construction builds on cluster-DAG identification [3] and is related to causal abstraction and macro-level structure learning [4, 8]. We coarsen manipulable inputs and explicitly characterize the actions removed by that choice.

7 Discussion and limitations

Exact invariance requires the correct quotient; quotient-visible errors and incorrect partitions remain harmful. The zero-price condition is sufficient and structural, and its cross-world consequence is not directly testable from single-world data. Refinement is heuristic, with recovery conditional on visiting a loss-less partition and on consistency of the base optimizer. Different partitions also induce different arm counts, initialization costs, and intervention costs. Finally, QMCBO inherits the causal-sufficiency assumption of its model-based factorization.

8 Conclusion

Coarsening makes precisely those graph distinctions that disappear in the quotient irrelevant to optimization. Within the resulting protected class, the complete trajectory is invariant. The price is the optimal value of fine interventions excluded by the parti-

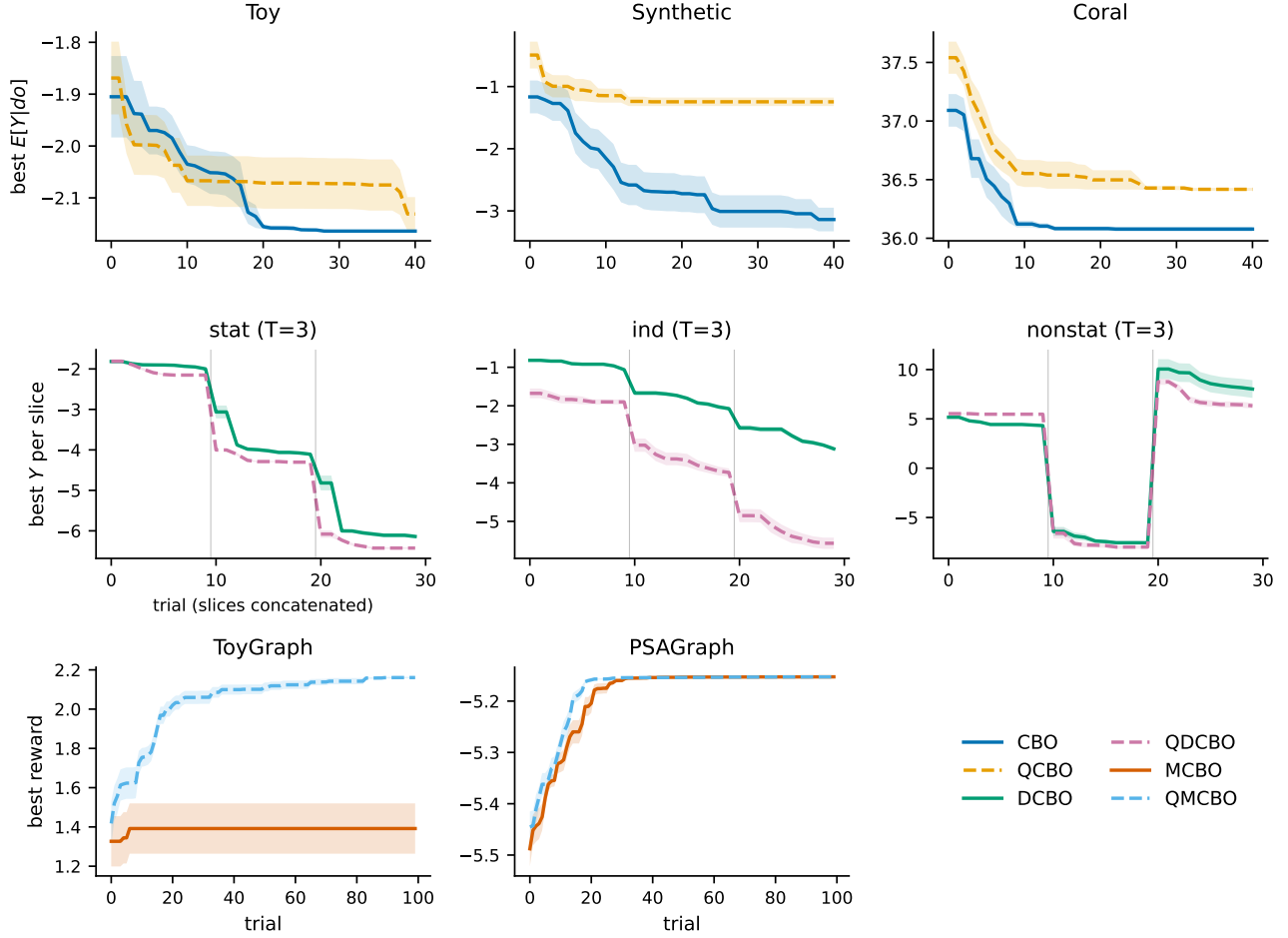


Figure 4: Quotient methods and their corresponding base methods under the released family protocols, reported as mean $\pm$ standard error. Top: CBO minimization. Middle: dynamic minimization over three time slices. Bottom: MCBO maximization.

tion, which is monotone under refinement and zero under the stated structural conditions. Theoretical and empirical results show that using quotients can prevent catastrophic graph errors, while hierarchical refinement provides a conditional route back to the fine action space.

References

- [1] V. Aglietti, X. Lu, A. Paleyes, and J. González. Causal Bayesian Optimization, May 2020.
- [2] V. Aglietti, N. Dhir, J. González, and T. Damoulas. Dynamic Causal Bayesian Optimization. In *Advances in Neural Information Processing Systems*, volume 34, pages 10549–10560. Curran Associates, Inc., 2021.
- [3] T. V. Anand, A. H. Ribeiro, J. Tian, and E. Bareinboim. Causal effect identification in cluster DAGs. <https://dl.acm.org/doi/10.1609/aaai.v37i10.26435>, 2023.
- [4] S. Beckers and J. Y. Halpern. Abstracting causal models. In *Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence and Thirty-First Innovative Applications of Artificial Intelligence Conference and Ninth AAAI Symposium on Educational Advances in Artificial Intelligence*, volume 33 of *AAAI’19/IAAI’19/EAAI’19*, pages 2678–2685, Honolulu, Hawaii, USA, Jan. 2019. AAAI Press. doi: 10.1609/aaai.v33i01.33012678.
- [5] N. Branchini, V. Aglietti, N. Dhir, and T. Damoulas. Causal Entropy Optimization. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, pages 8586–8605. PMLR, Apr. 2023.
- [6] S. Lee and E. Bareinboim. Structural Causal Bandits: Where to Intervene? *Advances in Neural Information Processing Systems*, 2018.

- [7] Y. Lu, A. Meisami, and A. Tewari. Causal Bandits with Unknown Graph Structure. In *Advances in Neural Information Processing Systems*, volume 34, pages 24817–24828. Curran Associates, Inc., 2021.
- [8] F. Madaleno, P. Misra, and A. Markham. Coarsening Causal DAG Models: Conference on Causal Learning and Reasoning (CLeaR). *Proceedings of the Fifth Conference on Causal Learning and Reasoning*, 2026.
- [9] S. Mukherjee, M. Zhang, S. Flaxman, and S. J. Vollmer. Graph Agnostic Causal Bayesian Optimisation, Nov. 2024.
- [10] J. Pearl. *Causality*. Cambridge University Press, Cambridge, 2 edition, 2009. doi: 10.1017/CBO9780511803161.
- [11] S. Sussex, A. Makarova, and A. Krause. Model-Based Causal Bayesian Optimization. 2023.

A Proofs and additional technical results

A.1 Method-specific results

Proposition 4 (Identity recovery). *At $\Pi = \Pi_{\text{fine}}$, the quotient graph equals the fine graph, the QCBO exploration set equals $\text{MIS}(\hat{\mathcal{G}})$, and its prior construction coincides arm by arm with CBO. QDCBO and QMCBO likewise reduce to DCBO and MCBO.*

Proof. At the singleton partition, χ is one-to-one and quotienting changes no edge. Unions of manipulable singleton clusters are exactly the subsets of $\mathbf{X}$, so Equation (5) is Equation (2). The C-DAG identification queries are then the same queries used by CBO. The dynamic slice graphs and model-based joint cluster mechanisms also reduce to their node-level counterparts. $\square$

Proposition 5 (Soundness of joint cluster mechanisms). *Under causal sufficiency and a valid partition, every intervention on a union of clusters S satisfies*

$$P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{C \notin S} P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)}) \Big|_{\mathbf{x}_S = \mathbf{x}_S}.$$

Proof. Apply the fine truncated factorization and group its factors by clusters. Within a cluster, the chain rule combines the member-level factors into the joint conditional mechanism. Validity orders the resulting mechanisms along the C-DAG. The joint conditional is essential: replacing it by a product of member-level marginals would discard within-cluster dependence. $\square$

Proposition 6 (Sub-cluster effects are not identified by cluster mechanisms). *There exist two causally sufficient SCMs with the same valid quotient, joint cluster mechanisms, and whole-cluster interventional distributions, but different values of $P(Y \mid \text{do}(A = a))$ for a strict subset A of a cluster.*

Proof. Let $C = \{X_2, X_3\}$ be a root cluster and $Y = X_2 + \varepsilon_Y$. In one SCM take $X_2 = \varepsilon_2$ and $X_3 = cX_2 + \varepsilon_3$; in another, choose the reverse linear-Gaussian factorization of the same joint distribution. The models agree on the joint law of C and every whole-cluster intervention. Under $\text{do}(X_3 = x)$, however, X_2 remains mean zero in the first model and changes linearly with x in the second. Quotient-level inputs therefore do not identify the sub-cluster effect. $\square$

A.2 Quotient invariance

Proof of Proposition 1. Write the run as decisions $d_1, d_2, \dots$ and datasets $\mathcal{D}_0, \mathcal{D}_1, \dots$, with

$$d_t = f_t(Q_\Pi(\hat{\mathcal{G}}), \mathcal{D}_{t-1}, \omega_t, \eta).$$

Equal quotients, initial data, random streams, and hyperparameters imply equal first decisions. The fixed true SCM produces the same coupled observation, so the updated datasets also agree. Induction gives equality of every decision, observation, incumbent, and stopping event. $\square$

Proof of Corollary 1. By construction, QCBO’s arm set and prior means depend on the supplied graph only through its C-DAG. QDCBO applies the same replacement within each time slice, and QMCBO composes only the joint mechanisms indexed by its C-DAG. Each construction therefore satisfies the quotient contract. Identity recovery follows from Proposition 4. $\square$

Proposition 7 (Necessity of the quotient contract). *If, for every ADMG $\mathcal{G}$, the trajectory map is constant on $E_\Pi(\mathcal{G})$ at fixed ξ , then it factors through Q_Π .*

Proof. For each quotient q , define $\bar{\tau}(q)$ as the common trajectory produced by any $\hat{\mathcal{G}}$ satisfying $Q_\Pi(\hat{\mathcal{G}}) = q$. Constancy over each protected class makes this definition independent of the representative. Hence

$$\tau(\hat{\mathcal{G}}) = \bar{\tau}(Q_\Pi(\hat{\mathcal{G}})).$$

$\square$

A.3 Price and protected-class separation

Lemma 1 (MIS reduction). *Let $\mathcal{G}$ be an ADMG with manipulable variables $\mathbf{X}$ and target Y . For $A \subseteq \mathbf{X}$, define*

$$A' := A \cap \text{An}_{\mathcal{G}_{\bar{A}}}(Y).$$

Then:

- (i) *for every $a \in D(A)$,*

$$P(Y \mid \text{do}(A = a)) = P(Y \mid \text{do}(A' = a|_{A'}));$$
- (ii) *if $A' \neq \emptyset$, then $A' \in \text{MIS}(\mathcal{G})$;*
- (iii) *consequently, optimizing over $\{\emptyset\} \cup \text{MIS}(\mathcal{G})$ attains the same population optimum as optimizing over all subsets of $\mathbf{X}$.*

Proof. Set $R := A \setminus A'$. Suppose that some $r \in R$ has a directed path to Y in $\mathcal{G}_{\bar{A}'}$. This path cannot enter A' . Let r^* be the last member of R on the path. The subpath from r^* to Y contains no further member of A , so it also exists in $\mathcal{G}_{\bar{A}}$. This contradicts $r^* \notin \text{An}_{\mathcal{G}_{\bar{A}}}(Y)$. Thus, while A' remains intervened on, variables in R cannot causally affect Y , proving (i).

For every $v \in A'$, a directed path from v to Y in $\mathcal{G}_{\bar{A}}$ cannot enter another member of A . It therefore also exists in $\mathcal{G}_{\bar{A}'}$, proving (ii). Applying (i) and (ii) to every candidate intervention proves (iii). $\square$

Consequently, when the null intervention is handled consistently, the best value over quotient MIS expanded to fine variables equals the best value over the corresponding cluster-union action family.

Proof of Proposition 2. Every union of clusters in Π is a union of clusters in a refinement Π' , and every such union is available at the singleton partition. The population minima are therefore over nested action families:

$$V(\Pi_{\text{fine}}) \leq V(\Pi') \leq V(\Pi).$$

Equality between the first and last terms holds exactly when a fine-optimal intervention belongs to $\mathcal{U}(\Pi)$. $\square$

Proof of Theorem 1. For part (i), use PARALLEL PARENT with manipulable variables $\{X_1, X_2\}$, domains $[-3, 3]^2$, edges $X_1 \rightarrow Y$ and $X_2 \rightarrow Y$, latent $X_1 \leftrightarrow X_2$, and partition $\{\{X_1, X_2\}, \{Y\}\}$. Choose the objective curvature so that the best $\{X_2\}$ -only intervention has value β , while the joint arm attains $y^* = 0$. Deleting $X_1 \rightarrow Y$ preserves the quotient through $X_2 \rightarrow Y$ but leaves $\text{MIS}(\hat{\mathcal{G}}) = \{\{X_2\}\}$. Every executed intervention and eligible incumbent therefore has regret at least β , giving $R_T \geq \beta T$. The whole-cluster arm remains globally optimal, so $\Delta(\Pi) = 0$.

For part (ii), Proposition 1 makes the trajectory and regret constant over $E_\Pi(\mathcal{G})$. Adding and subtracting $V(\Pi)$ gives

$$r_t = \Delta(\Pi) + \mu(\hat{z}_t) - V(\Pi).$$

The second term is nonnegative because every incumbent is an executed cluster-union action. Summing over t gives the stated decomposition. $\square$

A.4 Lossless coarsening

For the exact bridge used in this subsection, fix $A \subseteq \mathbf{X}$ and $a \in D(A)$. Let

$$B := A^\Pi \setminus A$$

be the variables added by cluster completion. Under $\text{do}(A = a)$, let $B(a)$ denote their natural response and $Y(a)$ the target response. Under $\text{do}(A = a, B = b)$, let $Y(a, b)$ denote the target response, and define

$$D_B(a) := \{b : (a, b) \in D(A^\Pi)\}.$$

Proposition 8 (Sufficient potential-outcome conditions for zero price). *If, for every $A \subseteq \mathbf{X}$ and $a \in D(A)$,*

$$\text{supp}(B(a)) \subseteq D_B(a) \quad \text{and} \quad B(a) \perp Y(a, b) \quad \text{for every } b \in D_B(a),$$

then $V(\Pi) = V(\Pi_{\text{fine}})$ and $\Delta(\Pi) = 0$.

Proof. Consistency and the stated independence give

$$\begin{aligned} \mathbb{E}[Y(a)] &= \int \mathbb{E}[Y(a, b) \mid B(a) = b] dP_{B(a)}(b) \\ &= \int \mathbb{E}[Y(a, b)] dP_{B(a)}(b) \\ &\geq \inf_{b \in D_B(a)} \mathbb{E}[Y(a, b)], \end{aligned}$$

where support inclusion justifies the final inequality. Thus $\mu^*(A^\Pi) \leq \mu^*(A)$. Taking minima and using $\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi_{\text{fine}})$ gives equality. $\square$

Lemma 2 (Cluster containment implies cross-world independence). *In the nonparametric structural equation representation of Definition 2, let $\mathbf{U}_B(A)$ be the exogenous ancestors of $A^\Pi \setminus A$ after $\text{do}(A)$, and let $\mathbf{U}_Y(A)$ be the exogenous ancestors of Y after $\text{do}(A^\Pi)$. Cluster-contained confounding implies*

$$B(a) \perp Y(a, b)$$

for every A , a , and $b \in D_B(a)$.

Proof. Recursive substitution writes

$$B(a) = f_B(\mathbf{U}_B(A); a), \quad Y(a, b) = f_Y(\mathbf{U}_Y(A); a, b).$$

Cluster-contained confounding requires $\mathbf{U}_B(A) \cap \mathbf{U}_Y(A) = \emptyset$. Because the exogenous variables are mutually independent, functions of the two disjoint sets are independent. $\square$

Proof of Corollary 2. Full intervention support gives $\text{supp}(B(a)) \subseteq D_B(a)$, while Lemma 2 gives $B(a) \perp Y(a, b)$. Proposition 8 then applies. $\square$

Remark 1 (Cross-world premise). *The independence $B(a) \perp Y(a, b)$ is not testable from a single observational or interventional distribution. It is not imposed as a primitive assumption in Corollary 2; it is derived from the structural condition in Definition 2.*

A.5 Refinement results

Proposition 9 (Refinement can invalidate a quotient). *There exist a valid partition Π and a refinement $\Pi' \preceq \Pi$ for which $\mathcal{G}^\Pi$ is acyclic but $\mathcal{G}^{\Pi'}$ contains a directed cycle. Validity must therefore be checked after every split.*

Proof. Let the fine graph contain $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$. The single cluster $\{a_1, a_2, b_1, b_2\}$ has an acyclic one-node quotient. Refining it into $\{a_1, a_2\}$ and $\{b_1, b_2\}$ exposes both quotient directions and produces a two-cycle. $\square$

Proof of Proposition 3. Every incumbent belongs to the fixed action family, so $\mu(\hat{z}_t) \geq V(\Pi)$. Subtracting $y^* = V(\Pi_{\text{fine}})$ gives $r_t \geq \Delta(\Pi)$; summing gives the cumulative bound. $\square$

Proof of Corollary 3. At a visited lossless partition, incumbent consistency gives $\mu(\hat{z}_t) \rightarrow V(\Pi_J) = y^*$, hence $r_t \rightarrow 0$ almost surely. Because R_T/T is the arithmetic mean of $r_1, \dots, r_T$, this also implies $R_T/T \rightarrow 0$. $\square$

Proof of Corollary 4. Proceed by induction over refinement phases. Within phase j , all graph-dependent behavior factors through $(\Pi_j, \mathcal{G}^{\Pi_j})$. Equal quotients and equal trajectory prefixes therefore produce the same phase trajectory, plateau decision, and next split. Repeating through Π_J proves equality of the adaptive trajectories. $\square$

B Acquisition and numerical estimation

B.1 Acquisition rules

After t evaluations of arm A , let $\mu_{A,t}(a)$ and $\sigma_{A,t}(a)$ be its GP posterior mean and standard deviation, and let y_t^{best} be the best observed interventional outcome. For minimization, expected improvement is

$$\text{EI}_{A,t}(a) := \mathbb{E} \left[(y_t^{\text{best}} - f_A(a))_+ \mid \mathcal{D}_t \right].$$

The arm-based implementation selects an arm and value maximizing $\text{EI}_{A,t}(a)/c(A,a)$, where $c(A,a)$ is the intervention cost. Model-based experiments retain their base method’s lower confidence bound for minimization, or upper confidence bound for maximization,

$$\mu_{A,t}(a) \mp \sqrt{\beta_t} \sigma_{A,t}(a),$$

where $\beta_t > 0$ controls exploration.

B.2 Causal-effect estimation

The complete identification check is performed on the ADMG supplied to the optimizer. When a query is identified, the returned functional is instantiated using the supported plug-in estimators.

For a valid backdoor set Z , we use

$$\widehat{\mathbb{E}}[Y \mid \text{do}(A = a)] = \int \widehat{\mathbb{E}}[Y \mid A = a, Z = z] d\widehat{P}(z).$$

For a frontdoor mediator M between treatment X and target Y , we use

$$\widehat{p}(y \mid \text{do}(x)) = \int \widehat{p}(m \mid x) \left\{ \int \widehat{p}(y \mid m, x') d\widehat{P}(x') \right\} dm.$$

Under causal sufficiency, the truncated factorization gives

$$\widehat{p}(v \setminus a \mid \text{do}(a)) = \prod_{V_i \in \mathbf{V} \setminus A} \widehat{p}(v_i \mid \text{pa}_i).$$

Complete identification determines whether a causal effect is a functional of the observational distribution, whereas backdoor, frontdoor, and g -computation describe the functionals currently implemented.

C Experimental details

Controlled suite. All methods receive the same sampled observational dataset and exact population intervention evaluator. Initial intervention values are sampled uniformly from the declared domains and their outcomes are computed by the same closed-form evaluator used during sequential optimization. Each run logs the selected arm and intervention values in addition to the incumbent. For protected perturbations, arm sequences and intervention values are compared exactly and incumbent values to 10^{-8} , allowing for numerical GP optimization. The complete perturbation taxonomy is reported in Table 3.

Table 3: Controlled misspecifications and measured paired trajectory deviations. Rows whose quotient is unchanged are predicted invariant by Proposition 1.

Cond.	Edit (assumed graph)	Quotient changed	Fine MIS changed	Fine prior changed	sup $ \Delta $
A1 (PP)	del $X_1 \rightarrow Y$ (quotient-redundant)	×	✓	—	0
A2 (PP)	add $X_1 \rightarrow X_2$ (intra-cluster bow)	×	×	✓	0
A3 (PP)	assume $X_1 \leftrightarrow Y$ (quotient-visible)	✓	×	✓	21.5
B1 (FD)	assume $X_1 \leftrightarrow M$ (intra-cluster conf.)	×	×	✓	0

Family suite. The CBO/QCBO comparison uses a common arm-based implementation with cost-normalized EI. DCBO/QDCBO and MCBO/QMCBO retain the released base implementations, including their acquisition and update schedules. All plots show means and standard errors over paired seeds.

D Dataset graphs and their quotients

For every distinct dataset topology, the following figures show the fine graph with its coarse partition and the quotient graph consumed by the optimizer. Manipulable variables are orange, targets are blue, dash-dotted edges denote latent confounding, and dashed boxes indicate manipulable clusters. For benchmarks whose graphs differ only by variable names and use the same partition structure, a single diagram is shown and its caption lists all corresponding benchmarks.



Figure 5: **ParallelParent (MinimalBench, Exp. A)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted edges denote latent confounding; dashed boxes are the manipulable clusters. The intra-cluster $X_1 \leftrightarrow X_2$ disappears in the quotient; deleting $X_1 \rightarrow Y$ leaves $C_1 \rightarrow Y$ alive through $X_2 \rightarrow Y$ (quotient-redundant).

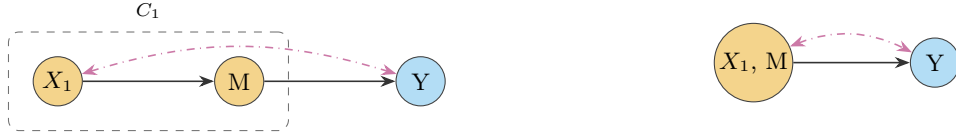


Figure 6: **FrontDoor (MinimalBench, Exp. B) / ToyGraph (CBO family)**. These datasets share the same fine and quotient topology under $X_1 \mapsto X$ and $M \mapsto Z$. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): the cluster arm uses the uninformative prior tier. In FRONTDOOR, the assumed intra-cluster $X_1 \leftrightarrow M$ therefore cannot change it.

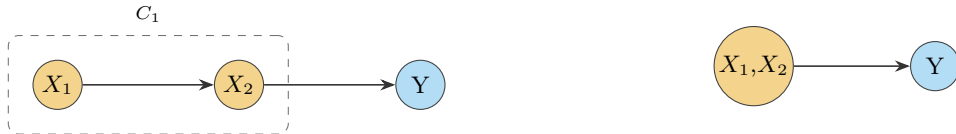


Figure 7: **MediatedChain (MinimalBench, Exp. C) / ToyGraph-M (QMCBO family)**. These datasets share the same chain-and-partition topology under $X_1 \mapsto X_0$ and $X_2 \mapsto X_1$. For MEDIATEDCHAIN, the policy box restricts $D(X_2)$ to $[-2, 2]$; the fine optimum $\text{do}(X_1=2)$ drives X_2 to 4, outside the box.

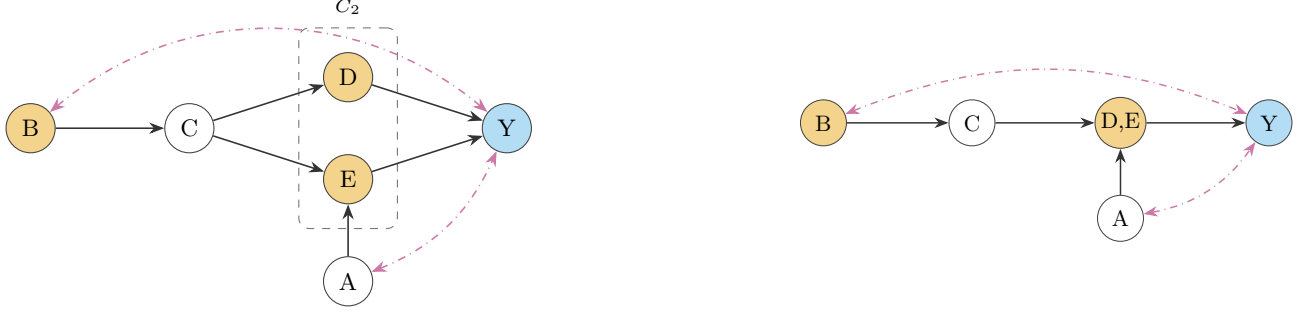


Figure 8: **CompleteGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

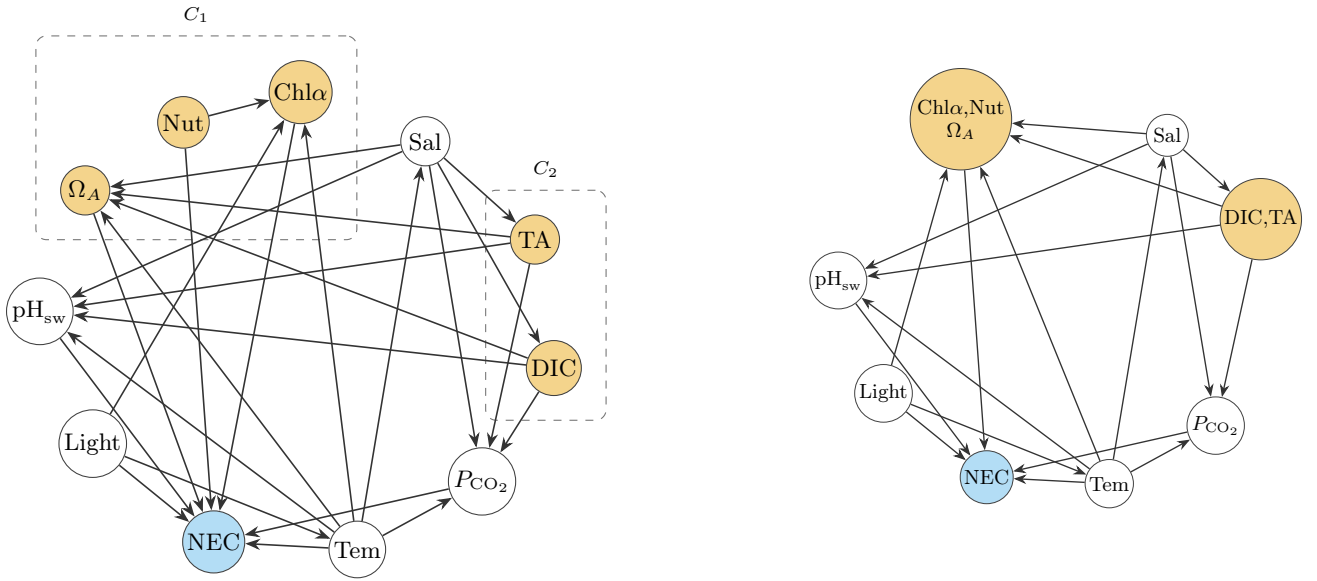


Figure 9: **SimplifiedCoralGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

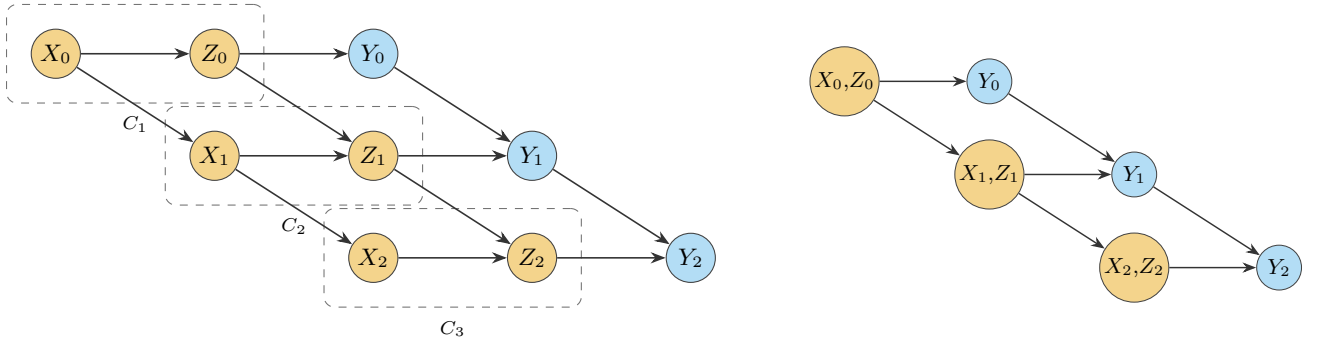


Figure 10: **DCBO stat / nonstat (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. Slices $t = 0, 1, 2$; the nonstat setup shares this topology with a change point in the SEM.

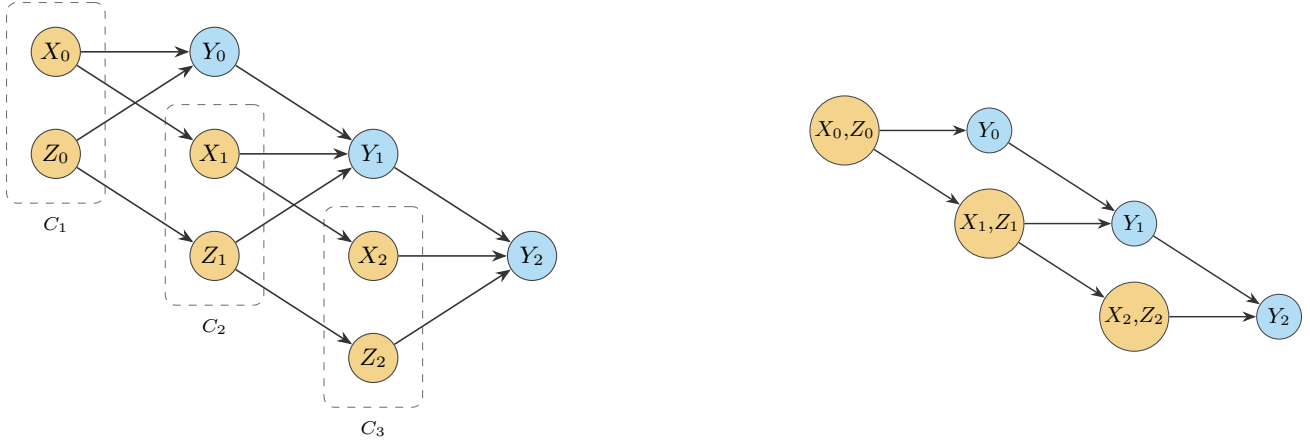


Figure 11: **DCBO ind (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

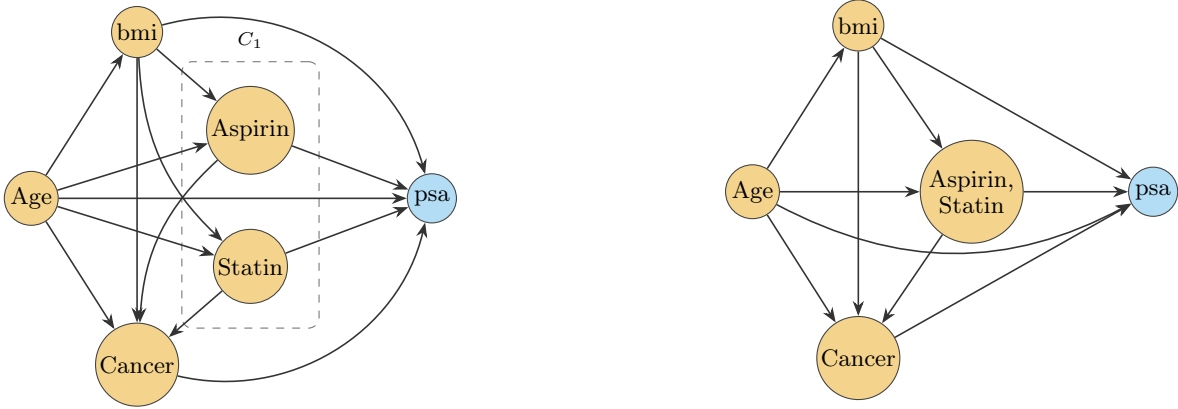


Figure 12: **PSAGraph (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.